



Switch Concepts

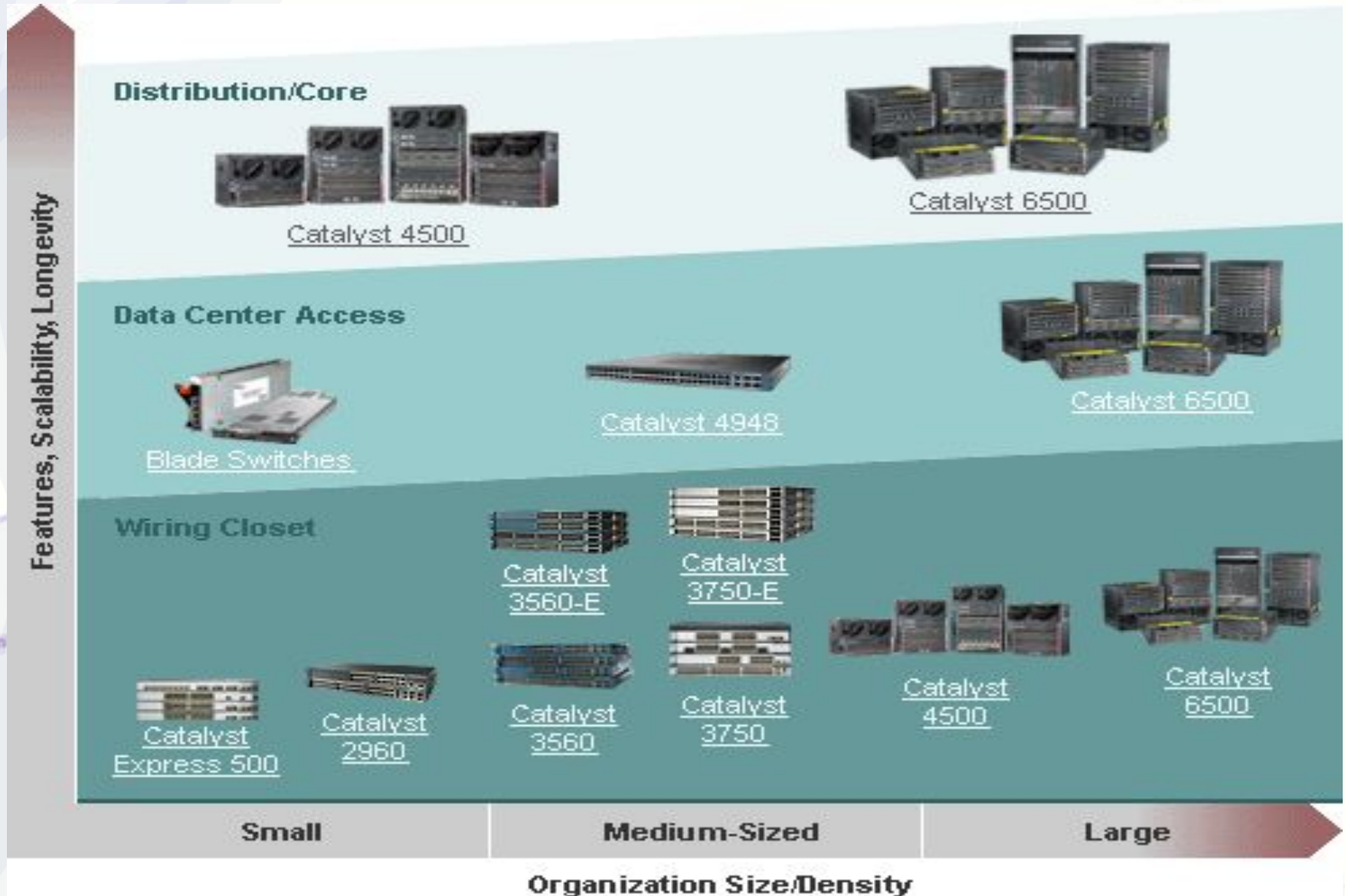
- ✓ Switch Configuration and Management
- ✓ Basic Switch Security Configuration
- ✓ **Types of Switches**
- ✓ Switch Architecture
- ✓ Switch Domains (Collision, Broadcast)

✓ Types of Networking Switches

Introduction

- Can your switches keep up with changing requirements?
- For networking builders, *network switch* is an essential component in their networking building plan. In a network deployment, switch channels incoming data from any of multiple input ports to the specific output port that will take the data toward its intended destination.
- Understanding the types of switches will help you find the right solution that's built for the future. There are categories of switches as well as specific switch benefits to consider as you explore your options.
- To achieve high performance level, there are different types of switches in networking.
- Ethernet network switches are broadly categorized into two main categories – *modular* and *fixed configuration*.
- There are variations to these types of network switches as switching is evolving, but the primary definitions remain the same.
- So, which is the most suitable one for your networking?

✓ Types of Networking Switches



✓ Types of Networking Switches

Modular Switches

- Modular switches as the name implies, allows you to add expansion modules into the switches as needed, thereby delivering the best flexibility to address changing networks. Examples of expansion modules are application-specific (such as firewall, wireless, or network analysis) and modules for additional interfaces, power supplies, or cooling fans.
- **Examples:** Facebook's "6-Pack" open modular switch platform, Cisco Catalyst 4K, 6K and 9K series.



Cisco Catalyst 9300 Series Switches
stacked using StackWise-480



Facebook's "6-Pack" open
modular switch platform

✓ Types of Networking Switches

Fixed Configuration Switches

- Fixed configuration switches are switches with a fixed number of ports and are typically not expandable.
- The fixed configuration switch category is further broken down into
 - unmanaged switches (*are basically plug and play*)
 - smart switches (*offer some management, QoS, and security*)
 - managed L2 and L3 switches (*high security, comprehensive features*)



unmanaged switches



smart switches



managed L2 & L3 switches

Can you spot the difference?

✓ Types of Networking Switches

Fixed Configuration Switches

- **Unmanaged Switches**
- An unmanaged switch is designed so that you can simply plug them in and they work, no configuration required. Unmanaged switches are typically for basic connectivity. You'll often see them used in home networks or wherever a few more ports are needed, such as at your desk, in a lab, or in a conference room.
- **Examples:** Netgear 5-port GB, Cisco 8-24port GB, D-Link 24 port GB, Tivega switches, TP-Link switches, etc.



✓ Types of Networking Switches

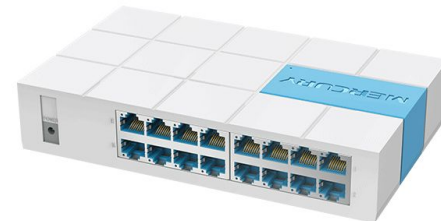
Fixed Configuration Switches

- **Unmanaged Switches ...cont.**
- This category of switch is the most cost effective where only basic layer 2 switching and connectivity is required. For example, they fit well when you need a few extra ports on your desk, in a lab, in a conference room, or even at home.
- With some unmanaged switches in the market, you can even get capabilities such as cable diagnostics, loop detection, prioritization of traffic using default QoS settings, Energy savings capabilities using EEE (Energy Efficient Ethernet) and even PoE (Power over Ethernet). However, as the name implies, these switches generally cannot be modified/managed. You simply plug them in and they require no configuration at all.

✓ Types of Networking Switches

Fixed Configuration Switches

- **Smart Switches**
- This category of switches is evolving. The general rule here is that these switches offer some management, QoS, and security, but they are “lighter” in capabilities and less scalable than managed switches. They can be a cost-effective alternative to managed switches.
- They can be deployed at the *edge of a large network* (with managed switches being used in the core), as the infrastructure for smaller networks, or for *low complexity needs*.
- **Examples:** Cisco Catalyst 250 Series, Smart Hub Desktop, TP-Link 8-port GB easy smart, NetGear Prosafe 5-Port, D-Link Web Smart 8-Port GB etc.



✓ Types of Networking Switches

Fixed Configuration Switches

- **Smart Switches...cont.**
- The capabilities available for this smart switch category vary widely. All of these devices have an interface for management that is typically more simplified than what managed switches offer.
- Smart switches allow you to segment the network into workgroups by creating VLANs, though with a lower number of VLANs and nodes (MAC addresses) than you'd get with a managed switch.
- They also offer some levels of security, such as 802.1x endpoint authentication, and in some cases with limited numbers of ACLs (access control lists), though the levels of control and granularity would not be the same as a managed switch.
- In addition, smart switches support basic quality-of-service (QoS) that facilitates prioritization of users and applications based on 802.1q/TOS/DSCP, adding to the versatility of the solution.

✓ Types of Networking Switches

Fixed Configuration Switches

- **Fully Managed L2 and L3 Smart Switches**
- The Managed switches are designed to deliver the most comprehensive set of features to provide the best application experience, the *highest levels of security*, the most *precise control and management* of the network, and *offer the greatest scalability* in the fixed configuration category of switches.
- **Examples:** D-Link 20-port Metro, Netgear Smart Managed Plus, Cisco Catalyst 1000 Series etc.



✓ Types of Networking Switches

Fixed Configuration Switches

- **Fully Managed L2 and L3 Smart Switches...cont.**
- As a result, managed switches are usually deployed as aggregation/access switches in very large networks or as core switches in relatively smaller networks. Managed switches should support both L2 switching and L3 IP routing though you'll find some with only L2 switching support.
- From a security perspective, managed switches provide protection of the data plane (User traffic being forwarded), control plane (traffic being communicated between networking devices to ensure user traffic goes to the right destination), and management plane (traffic used to manage the network or device itself). Managed switches also offer network storm control, denial-of-service protection, and much more.
- The Access Control List capabilities allows for flexibly dropping, rate limiting, mirroring, or logging of traffic by L2 address, L3 address, TCP/UDP port numbers, Ethernet type, ICMP or TCP flags, etc.

✓ Types of Networking Switches

Fixed Configuration Switches

- **Fully Managed L2 and L3 Smart Switches...cont.**
- Managed switches are rich in features that enable them to protect themselves and the network from deliberate or unintended Denial of Service attacks. It includes Dynamic ARP Inspection, IPv4 DHCP snooping, IPv6 First Hop Security with RA Guard, ND Inspection, Neighbor Binding Integrity, and much more.
- Additional security capabilities may include Private VLANs for securing communities of users or device isolation, Secure Management (downloads through SCP, Web-based Authentication, Radius/TACACS AAA, etc.), Control Plane Policing (CoPP) for protecting the CPU of the switch, richer support for 802.1x (time-based, Dynamic VLAN Assignment, port/host-based, etc)
- From a scalability perspective, these devices have large table sizes so that you can create large numbers of VLANs (for workgroups), devices (MAC table size), IP routes, and ACL policies for flow-based security/QoS purposes, etc.

✓ Types of Networking Switches

Fixed Configuration Switches

- **Fully Managed L2 and L3 Smart Switches...cont.**
- For highest network availability and uptime, managed switches support L3 redundancy using VRRP (Virtual Router Redundancy Protocol), large numbers of Link Aggregation groups (which is used both for scalability and resiliency), and capabilities for protecting L2 such as Spanning Tree Root Guard and BPDU Guard.
- For QoS and Multicast features, the richness of capabilities goes far beyond what is available in a smart switch. Managed switches support IGMP and MLD Snooping with functions for optimizing IPv4/v6 multicast traffic in the LAN, TCP Congestion Avoidance, 4 or 8 queues to treat traffic differently by importance, setting/tagging traffic by L2 (802.1p) or L3 (DSCP/TOS), and rate limiting traffic.

✓ Types of Networking Switches

Other Considerations

- Fixed configuration switches are switches with a fixed number of ports and are typically not expandable.
- In addition to the differences between switch categories, there are other options to consider, including: *network switch speeds, number of ports, Power over Ethernet, and stacking capabilities.*
- *Network switch speeds*
- Network switch speeds vary. You can find fixed configuration switches in Fast Ethernet (10/100 Mbps), Gigabit Ethernet (10/100/1000 Mbps), Ten Gigabit (10/100/1000/10000 Mbps) and even 40/100 Gbps speeds. Multigigabit technology is also available on some switches to deliver speeds beyond 1 Gigabit on existing Category 5e/6 cables. Switches have a number of uplink ports and a number of downlink ports. Downlinks connect to end users; uplinks connect to other switches or to the network infrastructure.

✓ Types of Networking Switches

Other Considerations...*cont*

- *Number of ports*
- Network switch sizes vary. Fixed configuration switches typically come in 5, 8, 10, 16, 24, 28, 48, and 52-port configurations. These ports may be a combination of SFP/SFP+ slots for fiber connectivity, but more commonly they are copper ports with RJ-45 connectors on the front, allowing for distances up to 100 meters. With Fiber SFP modules, you can go distances up to 40 kilometers. *SFP=Small Form Factor*
- *Power over Ethernet (PoE) versus non-PoE*
- POE is a capability that facilitates powering a device (such as an IP phone, IP Surveillance Camera, or Wireless Access Point) over the same cable as the data traffic. One of the advantages of PoE is the flexibility it provides in allowing you to easily place endpoints anywhere in the business, even places where it might be difficult to run a power outlet. One example is that you can place a Wireless Access Point inside a wall or ceiling.

✓ Types of Networking Switches

Other Considerations...*cont*

- *Power over Ethernet (PoE) versus non-PoE...cont.*
- Switches deliver power according to a few standards – IEEE 802.3af delivers power up to 15.4 Watts on a switch port whereas IEEE 802.3at (also known as PoE+) delivers power up to 30 Watts on a switch port. For most endpoints, 802.3af is sufficient but there are devices, such as Video phones or Access Points with multiple radios, which have higher power needs. Select Cisco switches also support Universal Power over Ethernet (UPoE) or 60W PoE that delivers up to 60 Watts on a switch port. A new PoE standard, 802.3bt, delivers even high levels of power for future applications.
- To find the switch that is right for you, choose a switch according to your power needs. When connecting to desktops or other types of devices which do not require PoE, the non-PoE switches are a more cost-effective option.

✓ Types of Networking Switches

Other Considerations...*cont*

- *Stackable vs. standalone switches*
- As a network grows, so does the number of switches. *Standalone switches* are managed and configured individually, while *stackable switches* are handled as a single unit. With a true stackable switch, eight switches (total 384 ports) function as a single switch – there is a *single SNMP/RMON agent, single Spanning Tree domain, single CLI or Web interface* – i.e. single management plane. You can also *create link aggregation groups spanning across multiple units in the stack, port mirror traffic from one unit in the stack to another, or setup ACLs/QoS spanning all the units.*
- Other advantages include *connecting the stack members in a ring* such that, if a port or cable fails, the stack will automatically route around that failure, many times at microsecond speeds. You *can also add or subtract stack members* and have automatic recognition and addition to the stack.
- **NB:** *Be careful about products in the market which are sold as “stackable” when they merely offer a single user interface, or central management interface, for getting to each individual switch unit. This approach is not stackable, but really “clustering”. You still have to configure every feature such as ACLs, QoS, Port mirroring, and more individually on each switch.*

Revision

- Which are the two main categories of ethernet switches? (1 mark).
- What is the main difference between *modular switches* and *fixed configuration switches*? (2 marks)
- Ethernet switches are designed to suite the different *sizes of organizations* i.e. *small, medium or large*. Which other main factors, regardless of size can be considered? (3 marks). *ans: features, scalability, longevity, security, management etc.*
- Fixed Configuration switches can be put into 3 main categories. Explain each briefly. (3 marks).
- What's the key characteristic of *unmanaged switches*? (1 mark).
- Managed switches are designed to deliver the most comprehensive set of features. Which are some of these features? Name three. (3 marks)
- What is one main difference in configuration when you have real *stackable switches* and when you have standalone *clustered switches*? i.e. *Switch Stack vs. Switch Cluster*.



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